

The Liquid Galactica

Toward Persistent Digital Minds: Whole-Brain Emulation, Causal Information,
Quantum-Biology-Aware Modeling, and AI-Accelerated Neurocomputing

Devanik Debnath (Devanik21)

Electronics and Communication Engineering student, National Institute of Technology, India
Independent technical researcher

September 20, 2026

Final technical preprint / quantitative research framework

Abstract

The human brain is a biological information-processing system with an unusual combination of scale, plasticity, robustness, and energy efficiency. Contemporary estimates place its power consumption near 20 W, while recent connectomic experiments have produced petascale reconstructions from cubic-millimetre volumes and large-scale simulations have reached the approximate neuron count of the human brain. In parallel, frontier artificial intelligence is becoming increasingly capable of long-horizon coding, scientific reasoning, automated experimentation, and algorithm discovery. These trajectories motivate a long-term research question: whether an individual human mind could eventually be represented, reconstructed, and persistently executed on a non-biological substrate.

This paper introduces *Liquid Galactica*, a research framework for that problem. The objective is intentionally narrower than a claim that mind uploading is imminent. We seek to formalize the information, computation, energy, measurement, and validation requirements for a future whole-brain emulation (WBE) system, and to identify technologies that could reduce those requirements. The central proposal is to treat a brain not as a static connectome but as a multiscale, time-dependent causal state whose sufficient representation must preserve the observables and intervention responses relevant to cognition and memory. We formulate a *Minimum Sufficient Causal Brain State* problem using information theory, dynamical systems, system identification, and rate-distortion ideas.

We additionally introduce a quantum-biology-aware branch of the framework. This branch does *not* assume that consciousness is a quantum phenomenon. Instead, it asks whether quantum effects known to occur in biological chemistry, together with quantum-information formalisms for open systems, can supply useful microscopic models, compression tools, or falsifiable tests for future emulation. Candidate formalisms include density operators, quantum channels, Lindblad dynamics, tensor-network factorizations, and quantum fidelity, all coupled to classical membrane, synaptic, and network dynamics.

Finally, we place WBE on a technology-coupled timeline. As of September 2026, public frontier evidence includes increasingly capable agentic models, AI systems used for scientific research and algorithm discovery, GPT-6 Astra and Claude Fable 5.1, and Google’s confirmed pre-training of Gemini 4. Contemporary forecasting systems remain uncertain, but a working optimistic scenario places broadly general high-level AI in the late 2020s to early 2030s, with a wider and more uncertain interval for superintelligence. Liquid Galactica treats this acceleration as a multiplier on neuroscience and engineering research rather than a guarantee. The resulting program is a proposal for measurement, reconstruction, compression, validation, and eventually persistent digital instantiation of human neural organization.

Keywords: whole-brain emulation; connectomics; neural dynamics; memory engrams; causal representation; information theory; neuromorphic computing; quantum biology; quantum information; consciousness; digital persistence; recursive self-improvement; AI for science.

Author note. Devanik Debnath (Devanik21) is an Electronics and Communication Engineering student at a National Institute of Technology in India and an independent technical researcher. His public technical portfolio is maintained under the identifier “Devanik21” at <https://github.com/Devanik21>. This profile is included for authorship context and provenance only; it is not scientific evidence for any claim in this manuscript. **Literature cut-off.** This final preprint uses information publicly available on or before September 20, 2026. Current frontier-AI claims are attributed to reporting laboratories or forecasting sources; biological and quantum claims are explicitly separated into observation,

inference, projection, and hypothesis.

1. Scope and research posture

The title *Liquid Galactica* denotes a research direction rather than a finished technology. The word “liquid” is used to emphasize that the relevant object may be a continuously evolving information process rather than a frozen file, while “Galactica” denotes the scale of the proposed systems problem: a human mind must be represented across many spatial, temporal, biochemical, and computational orders of magnitude.

The present manuscript is a technical framework and

literature synthesis. It contains no new biological measurements, no claim of successful mind uploading, and no assertion that subjective consciousness has been reduced to a known algorithm. Wherever an idea extends beyond established experimental evidence, it is marked as a hypothesis, a candidate formalism, or an engineering target. This distinction is essential because the problem combines mature subfields with questions that remain unresolved even within contemporary neuroscience.

Whole-brain emulation was articulated as a roadmap more than a decade ago by Sandberg and Bostrom, who decomposed the challenge into scanning, image processing, reconstruction, parameter estimation, simulation, and validation (Sandberg and Bostrom, 2008). The field now contains substantially stronger empirical demonstrations at several layers of this pipeline, but the central gap remains: no individual human brain has been measured, reconstructed, and validated at the multiscale resolution needed for a demonstrated personal emulation. The 2025 State of Brain Emulation Report presents this gap explicitly and frames the field around recording, wiring reconstruction, and digital modelling (Zanichelli et al., 2025).

The present work therefore adopts the following operational objective:

Determine the smallest experimentally grounded representation of an individual human brain that preserves the causal dynamics required for a specified class of cognitive, mnemonic, behavioral, and potentially conscious functions.

This wording is intentionally compatible with several eventual outcomes. A future reconstruction might preserve behavior but not consciousness; it might preserve functional dynamics without establishing personal continuity; or it might eventually support a stronger notion of digital persistence. The framework does not pre-judge those outcomes.

2. The physical information problem

2.1 From tissue to data

The first quantitative observation is simple: biological tissue can encode an enormous amount of experimentally accessible structure. Shapson-Coe et al. reconstructed a petavoxel-scale fragment of human temporal cortex at nanoscale resolution. The publicly discussed reconstruction comprised roughly 1.4 petabytes of imagery and included tens of thousands of cells and approximately 150 million synapses in a cubic-millimetre-scale volume (Shapson-Coe et al., 2024).

A naive extrapolation suggests that whole-brain raw image archives can move from exabyte to zettabyte scales as resolution approaches tens of nanometres. The 2025 State of Brain Emulation Report estimates approximately 128–180 EB for a 25-nm isotropic whole-human-brain imaging regime and approximately 2–2.8 ZB for a 10-nm regime, under its stated assumptions (Zanichelli et al., 2025). These values are not direct estimates of the information required to execute a mind. They are

estimates of a high-volume measurement representation.

Let brain volume be V , isotropic spatial resolution be r , and effective bytes per voxel be b . A first-order raw-data estimate is

$$D_{\text{raw}} \approx Vr^{-3}b. \quad (1)$$

Equation (1) immediately motivates a central design objective. If an image archive contains substantial redundancy, then storing the archive is a sufficient condition for preserving information but not necessarily a minimal condition. We therefore define a representation map

$$\Phi : I_{\text{raw}} \mapsto R, \quad (2)$$

where I_{raw} is the measured data and R is a structured neural representation.

A successful compressor must be constrained by downstream causal behavior. A visually accurate segmentation that changes an important synaptic motif, membrane parameter, or neuromodulatory state may be unacceptable even if conventional pixel-level reconstruction error is low. This motivates a distinction between *data compression* and *causal compression*.

2.2 A multiscale brain state

We model an individual brain at time t by

$$\mathcal{B}(t) = (C, N, S, V, P, M, G, X, E, U)_t, \quad (3)$$

where C denotes connectivity, N neuronal identities and morphology, S synaptic parameters, V electrical state, P plasticity variables, M neuromodulatory state, G glial state, X intracellular and molecular state, E extracellular and vascular context, and U embodied sensory and motor coupling.

This decomposition is not asserted to be a unique ontology. It is a convenient partition for engineering. The real scientific question is whether there exists a smaller state sufficient for the target task. We therefore define

$$R^*(\varepsilon, \mathcal{T}) = \arg \min_R |\mathcal{R}| \quad \text{s.t.} \quad D_{\mathcal{T}}(\mathcal{B}, R) \leq \varepsilon, \quad (4)$$

where $\mathcal{T}$ is a class of tests and $D_{\mathcal{T}}$ is a task-conditioned causal distortion measure.

The notation is deliberately general. If the target is only visual discrimination, a very compact model might suffice. If the target is autobiographical memory, lifelong learning, intervention response, and open-ended interaction, the required representation may become much richer.

2.3 Static structure versus dynamic state

A static connectome is a graph

$$G = (\mathcal{V}, \mathcal{E}), \quad (5)$$

with neurons represented as vertices and synaptic relationships as edges. But biological computation is not defined by graph topology alone. Synapses have strengths and molecular states; neurons carry membrane potentials and conductance variables; plasticity modifies the graph over time; glia and neuromodulators alter

effective computation; and the system is continuously driven by body and environment.

The dynamical model can therefore be written abstractly as

$$\dot{z}(t) = F(z(t), \theta(t), u(t)), \quad y(t) = H(z(t)), \quad (6)$$

where $z(t)$ is the latent neural state, $\theta(t)$ is a time-varying parameter set, $u(t)$ is input from body and environment, and $y(t)$ is an observable output.

The reconstruction problem becomes a system-identification problem under incomplete observation:

$$\hat{F}, \hat{\theta}, \hat{z}_0 = \arg \min_{F, \theta, z_0} \mathcal{L}(y_{1:T}, \hat{y}_{1:T}) + \lambda \Omega(F, \theta), \quad (7)$$

subject to biological constraints and uncertainty intervals.

3. Connectomics at the edge of the possible

3.1 Human cortex: petascale evidence

The H01 human cortex reconstruction demonstrates that a millimetre-scale piece of human brain can already become a massive digital object (Shapson-Coe et al., 2024). It also demonstrates the importance of algorithmic reconstruction: the scientific value is not the raw pixels alone, but the segmentation, neurite tracing, synapse detection, morphology, and browsable representations extracted from them.

This observation suggests a future “neural compiler” pipeline:

$$\begin{aligned} &\text{microscopy} \rightarrow \text{segmentation} \rightarrow \text{proofreading} \\ &\rightarrow \text{graph construction} \rightarrow \text{parameter inference} \\ &\rightarrow \text{validated emulator.} \end{aligned} \quad (8)$$

The pipeline must be error-aware. Let p_e denote an estimated probability of structural error on an inferred edge. For a graph with N_e relevant edges, the expected number of errors is approximately

$$\mathbb{E}[K] \approx N_e p_e. \quad (9)$$

When N_e becomes enormous, very small local error probabilities can generate large absolute error counts. The objective therefore cannot be “near-perfect imaging” in an informal sense; it must become a calibrated measurement problem with explicit uncertainty propagation and downstream sensitivity analysis.

3.2 Mouse functional connectomics

The MICrONS dataset is a particularly important bridge between structure and dynamics. Dense calcium imaging captured roughly 75,000 neurons in an awake mouse across multiple visual areas, while electron microscopy reconstructed more than 200,000 cells and approximately 0.5 billion synapses in the corresponding volume. The datasets were co-registered, enabling analysis that joins neural activity, morphology, connectivity, cell type, and gene-expression-associated information (MICrONS Consortium, 2025).

The significance for Liquid Galactica is methodological. The future brain-state representation should ideally

be learned from co-registered modalities rather than treating anatomy, activity, molecular identity, and behavior as independent datasets.

3.3 The fly and the lesson of completeness

The complete adult *Drosophila* brain connectome contains approximately 139,000 neurons and more than 50 million synaptic connections, demonstrating that complete structural reconstruction is possible for an organism dramatically smaller than a human (Dorkenwald et al., 2024). The transition from such systems to a human brain is not a simple linear scaling problem. The number of possible cell types, the complexity of long-range connectivity, the molecular state space, and the diversity of temporal dynamics all expand simultaneously.

A useful scaling heuristic is

$$\mathcal{C}_{\text{WBE}} \sim N_n \bar{d} K_s R_t M_b, \quad (10)$$

where N_n is neuron count, $\bar{d}$ average relevant degree, K_s state dimensionality per synapse or cell, R_t temporal sampling burden, and M_b a model-breadth factor capturing biological detail. This is not a physical law. Its purpose is to show why “number of neurons” alone is an inadequate complexity measure.

4. Memory as a distributed physical process

A useful individual representation must preserve the information that makes the individual unlike an arbitrary brain-sized network. Memory is therefore one of the most demanding requirements.

A 2025 survey of 312 neuroscientists found that 70.5% agreed long-term memories are primarily maintained by neuronal connectivity patterns and synaptic strengths, but there was no clear consensus on which specific neurophysiological features or scales are essential. The same survey gave a median estimate of roughly 40% for whether a static structural snapshot could yield some long-term memories and reported very broad forecasts for future WBE feasibility (Zelevnikov-Johnston et al., 2025).

This uncertainty motivates an engram-aware state model:

$$M_k(t) = \mathcal{F}_k(C(t), S(t), V(t), P(t), X(t), u_{0:t}), \quad (11)$$

where M_k denotes an individual memory or memory ensemble. The function $\mathcal{F}_k$ is not known. The equation simply encodes a research hypothesis: a memory may be represented by a distributed trajectory through structural, electrical, and molecular state spaces rather than by a localized record.

4.1 Memory fidelity as a testable hierarchy

We propose four nested memory criteria:

1. **Recognition fidelity:** the system recognizes previously encountered objects, places, people, or concepts.
2. **Recall fidelity:** the system can retrieve relevant information with similar content and temporal organization.

3. **Associative fidelity:** the same cues produce comparable associations and cross-links.
4. **Plasticity fidelity:** the memory participates in future learning, reconsolidation, interference, and generalization in a biologically comparable way.

The fourth criterion is particularly important. A static transcript of a memory may fail to preserve the dynamics that allow the memory to change.

5. Information theory of mind reconstruction

5.1 Causal rate-distortion

Classical rate-distortion theory seeks a representation of a source B that minimizes mutual information subject to an acceptable distortion. For a brain, a standard Euclidean distortion is not meaningful. We propose a task-conditioned causal distortion:

$$D_c(B, R) = \mathbb{E}_{I \sim \mathcal{I}} [W(P_B(Y \mid \text{do}(I)), P_R(Y \mid \text{do}(I)))], \quad (12)$$

where $\mathcal{I}$ is a set of admissible interventions, Y denotes measured outcomes, and W is a distance between outcome distributions.

The corresponding causal rate-distortion function is

$$R_c(\varepsilon) = \min_{p(r|b)} I(B; R) \quad \text{s.t.} \quad D_c(B, R) \leq \varepsilon. \quad (13)$$

This is proposed as a research formalism, not as an established neuroscience result. Its value is conceptual: it changes the compression objective from “preserve what the image looks like” to “preserve what the system can causally do.”

5.2 Minimum sufficient causal representation

A representation R is *task-sufficient* if the conditional distribution of target observables is approximately preserved:

$$P_B(Y_{1:T} \mid U_{1:T}) \approx P_R(Y_{1:T} \mid U_{1:T}). \quad (14)$$

For intervention-aware sufficiency, we require the approximation to hold under a family of interventions $\mathcal{I}$:

$$\sup_{i \in \mathcal{I}} D_W(P_B(Y \mid \text{do}(i)), P_R(Y \mid \text{do}(i))) \leq \varepsilon. \quad (15)$$

This definition is useful because it admits progressive emulation. Early systems can target narrow task classes. As the task set expands from perception to memory, embodiment, language, and open-ended learning, the representation can be refined.

5.3 Kolmogorov-style intuition

One may also consider a description-length objective:

$$L(R) + L(\mathcal{M} \mid R), \quad (16)$$

where $L(R)$ is the description length of the representation and $L(\mathcal{M} \mid R)$ measures the residual complexity of the model needed to generate its observed dynamics. The goal is not necessarily to minimize $L(R)$ alone. A representation can be tiny but require a massive hidden simulator. The meaningful quantity is therefore total executable description length under a stated physical substrate.

6. Compute: from neurons to executable dynamics

6.1 Brain-scale simulation already exists in approximate form

Lu et al. demonstrated a large-scale digital-brain system with up to 86 billion model neurons and 47.8 trillion neuronal synapses, using 14,012 GPUs. The model was constrained by personalized MRI information and biological priors and reproduced selected resting-state and visual-response properties (Lu et al., 2024).

This result is important precisely because it is not an uploaded person. It establishes that human-neuron-count-scale simulation can be implemented in a practical cluster architecture, while leaving open the much harder question of individualized parameter identification.

6.2 FLOPS are not a sufficient brain metric

A single floating-point operation is not a biologically privileged unit. Different neuron models have different state dimensionality, communication costs, and numerical requirements. A useful emulator could operate with event-driven rules, reduced-order models, lookup tables, analog dynamics, or custom accelerators.

We therefore define an emulation cost vector

$$\begin{aligned} \mathbf{K} &= (K_{\text{state}}, K_{\text{events}}, K_{\text{memory}}), \\ \mathbf{K}' &= (K_{\text{bandwidth}}, K_{\text{latency}}, K_{\text{energy}}). \end{aligned} \quad (17)$$

The optimization target becomes multi-objective:

$$\min \alpha_1 K_{\text{energy}} + \alpha_2 K_{\text{latency}} + \alpha_3 K_{\text{memory}} + \alpha_4 K_{\text{bandwidth}} \quad (18)$$

subject to fidelity constraints.

7. The 20 W benchmark and the energy problem

The human brain is widely reported to operate near a 20-W power budget. Recent reviews continue to use this figure as a benchmark for brain-inspired computing (Chiel et al., 2026; Bruno et al., 2026). The value should not be interpreted as a single “20-W computation” because energy is distributed across electrical signaling, ion pumping, synaptic transmission, cellular maintenance, vascular support, and other processes.

This nevertheless presents a remarkable engineering target. A future digital mind need not exactly reproduce biological energy consumption, but approaching it would make persistent large-scale emulations dramatically more attractive.

The thermodynamic lower bound for irreversible bit erasure is

$$E_{\text{Landauer}} = k_B T \ln 2. \quad (19)$$

At $T \approx 310$ K, this is approximately 3×10^{-21} J per erased bit. This bound is a floor, not a prediction of brain energy efficiency. Actual biological computation operates many orders of magnitude above the bound because of reliability, finite-time dynamics, chemical overhead, transport, communication, and physical constraints.

A useful long-term metric for Liquid Galactica is

therefore

$$\eta_c = \frac{\text{validated causal information processed}}{\text{joule}}, \quad (20)$$

where the numerator must be defined by a reproducible task suite rather than by an ambiguous FLOP count.

7.1 Communication may dominate

Brain-inspired hardware literature repeatedly emphasizes that data movement between memory and compute can be a major cost in conventional systems. Biological synapses colocate storage and communication with nonlinear computation, suggesting that a future WBE substrate may benefit from event-driven in-memory or near-memory architectures (Bruno et al., 2026; Chiel et al., 2026).

A hypothetical architecture could therefore follow

$$\text{storage} \approx \text{state} \approx \text{compute} \quad (21)$$

rather than the classical separation of memory and processor.

8. A hybrid computational substrate

Liquid Galactica proposes a heterogeneous execution stack rather than a monolithic simulator:

1. a molecular layer for selected biochemical dynamics;
2. a cellular layer for membrane and intracellular state;
3. a synaptic layer for plasticity and transmission;
4. a network layer for spikes, oscillations, and recurrent interactions;
5. a cognitive layer for higher-order state variables and memory structures;
6. an embodied layer coupling the model to sensory and motor interfaces.

Each layer can use a different computational abstraction. There is no scientific requirement that every molecule be simulated by first-principles quantum mechanics. The engineering requirement is that omitted degrees of freedom be compressed without violating the fidelity criterion relevant to the target behavior.

9. Quantum biology: established phenomena and disciplined extrapolation

Quantum mechanics is already part of biology in several experimentally supported contexts. Reviews of quantum biology discuss electron transfer, enzyme reactions, proton tunneling, photosynthetic energy transport, spin-dependent chemistry, ion channels, and magnetoreception (Lambert et al., 2021). In migratory animals, radical-pair chemistry involving cryptochrome proteins is an important leading mechanism for magnetoreception, although the exact biological implementation remains actively studied (Hore and Mouritsen, 2016; Kim et al., 2023).

These facts justify a careful quantum-biology component in a WBE framework. They do *not* establish that human cognition is a quantum computer or that consciousness depends on long-lived neuronal entanglement.

9.1 Open quantum systems as a modeling layer

If a future measurement program identifies a biologically relevant quantum subsystem, its reduced state can be represented by a density operator ρ satisfying

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H, \rho] + \sum_j \left(L_j \rho L_j^\dagger - \frac{1}{2}\{L_j^\dagger L_j, \rho\} \right), \quad (22)$$

where H describes coherent dynamics and L_j represent dissipative channels.

Equation (22) is a generic open-system model. The point is methodological: if quantum degrees of freedom are demonstrated at a relevant scale, the WBE representation should be capable of representing them rather than silently forcing every phenomenon into a classical approximation.

9.2 Quantum fidelity as a preservation criterion

For candidate quantum states ρ and σ , Uhlmann fidelity provides a natural closeness measure:

$$F(\rho, \sigma) = \left(\text{Tr} \sqrt{\sqrt{\rho} \sigma \sqrt{\rho}} \right)^2. \quad (23)$$

A future quantum-aware brain emulator could therefore be evaluated by a hybrid metric

$$D_{\text{hybrid}} = \lambda_c D_c + \lambda_q (1 - F_Q) + \lambda_e D_E, \quad (24)$$

where D_c measures classical causal mismatch, F_Q quantum-state fidelity on experimentally validated subsystems, and D_E measures energy or resource mismatch.

9.3 Quantum consciousness: hypothesis, not premise

Orchestrated Objective Reduction (Orch-OR) is one of the best-known hypotheses proposing an important role for coherent quantum processes in neuronal microtubules (Hameroff and Penrose, 2014). Recent papers continue to argue for quantum microtubule mechanisms, while broader reviews of consciousness remain pluralistic and do not treat a quantum mechanism as established consensus (Hameroff, 2025; Consciousness perspective, 2024; Namazi, 2026).

Liquid Galactica therefore treats quantum consciousness as a falsifiable branch of the research tree:

If future experiments demonstrate that a specific quantum degree of freedom is necessary for a target conscious function, the emulator must preserve that degree of freedom or an experimentally validated computationally equivalent representation.

No stronger assumption is required.

10. Tensor networks and quantum-inspired compression

Tensor networks offer a second route to “exotic” mathematics without requiring the brain itself to be quantum. High-dimensional systems often contain correlations that can be represented compactly when the relevant structure is low-rank or approximately factorizable.

Let a high-order tensor $T_{i_1 \dots i_n}$ represent a discretized multiscale state. A tensor-train approximation seeks factors

$$T_{i_1 \dots i_n} \approx G_1^{i_1} G_2^{i_2} \dots G_n^{i_n}, \quad (25)$$

with intermediate ranks controlling complexity.

For Liquid Galactica, the research question is not “is the brain a tensor network?” It is:

$$\boxed{\exists R_{\text{low}} : D_c(\mathcal{B}, R_{\text{low}}) \leq \varepsilon ?} \quad (26)$$

A successful answer would provide a principled compression mechanism. A negative answer would also be useful because it would identify where irreducible high-order correlations prevent strong compression.

10.1 Fractal and multiscale structure

The brain exhibits structure over scales ranging from molecular assemblies to whole-network anatomy. A multiscale representation can therefore be organized as

$$\mathcal{B}_{\ell+1} = \mathcal{A}_\ell(\mathcal{B}_\ell), \quad (27)$$

where $\mathcal{A}_\ell$ is a learned or mechanistic coarse-graining map between scales.

One research program is to search for approximately scale-invariant statistics, sparse factorization, graph motifs, and renormalization-like transformations that reduce simulation burden while preserving observables. Such an approach would borrow mathematical tools from statistical physics without claiming that the brain literally realizes a particular field-theoretic universality class.

11. A causal brain codec

We propose a conceptual system called the *Causal Brain Codec* (CBC). The codec takes multimodal observations and produces an executable representation with explicit uncertainty.

Let observed data be D and latent brain representation be R . The codec estimates

$$p(R | D) \propto p(D | R)p(R). \quad (28)$$

Instead of producing a single deterministic reconstruction, it produces a posterior or ensemble. This is important because uncertainty will remain at multiple layers: missed synapses, ambiguous cell boundaries, unknown molecular parameters, and incomplete state recordings.

The executable model then propagates uncertainty:

$$p(z_{t+1} | z_t) = \int p(z_{t+1} | z_t, \theta) p(\theta | D) d\theta. \quad (29)$$

A future system could therefore report not merely “this neuron fires at 17 Hz” but a calibrated distribution over firing dynamics under uncertainty in the inferred state.

12. AI-accelerated neuroscience

12.1 The transition from AI assistant to AI research participant

The technological landscape changed materially during 2025–2026. Google’s AI co-scientist was designed as a

multi-agent system for generating hypotheses, research plans, and experimental proposals, with iterative generation, reflection, ranking, and evolution loops (Gottweis et al., 2025). The *AI Scientist* program demonstrated an end-to-end system that generates ideas, writes code, executes experiments, analyzes results, writes manuscripts, and performs automated review; a later Nature paper reported that the system’s generated manuscript passed the first round of peer review at a workshop of a top-tier machine-learning conference (Lu et al., 2026).

Google’s AlphaEvolve pushed algorithm discovery further by combining language-model generation with automated evaluators and evolutionary search. Google reported applications in mathematics, computing infrastructure, and scientific workloads (AlphaEvolve team, 2026).

These systems do not establish autonomous recursive intelligence in the strong sense. They do demonstrate that portions of the research loop can already be delegated to software.

12.2 Recursive self-improvement as a research variable

A September 2026 preprint, *The Last AI Built by Humans: Toward Genuine Recursive Self-Improvement*, formalizes a roadmap from improvement-execution autonomy through improvement-strategy, experience acquisition, environment adaptation, and recursive meta-improvement, while explicitly discussing the practical gaps that remain (Duan et al., 2026).

This paper adopts a conservative interpretation. We define an RSI acceleration factor

$$A_{\text{RSI}}(t) = \frac{\text{effective AI-R\&D throughput at } t}{\text{effective AI-R\&D throughput at baseline}}, \quad (30)$$

where throughput may include algorithm design, experiment generation, code implementation, evaluation, and model analysis.

The important point for Liquid Galactica is that A_{RSI} can multiply progress in other technologies. If AI systems help design microscopes, interpret images, discover compression algorithms, fit biophysical parameters, optimize neuromorphic hardware, and accelerate molecular biology, the brain-emulation roadmap becomes coupled to the AI research loop.

13. Late-2026 frontier and the AGI/RSI/ASI scenario envelope

The late-2026 AI landscape is unusually important to the Liquid Galactica program because AI capability is itself a possible accelerator of every other workstream. OpenAI reports GPT-6 Astra as a frontier model spanning computer use, software engineering, science, cybersecurity, and mathematical reasoning; it reports 98% on FrontierMath Tier 4 and 99.9% on ARC-AGI-3 (OpenAI, 2026). Anthropic’s Fable 5.1 and Mythos 5.1 are positioned for advanced coding, knowledge work, and research workflows (Anthropic, 2026). Google has publicly stated that its Gemini 4 pre-training run is its most ambitious yet; this is evidence of continued frontier-model development, not evidence about unreleased Gemini 4

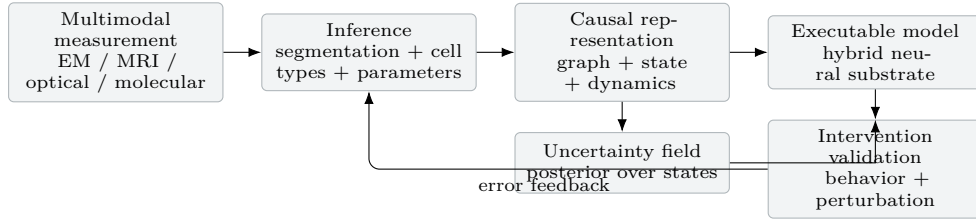


Figure 1: The proposed Causal Brain Codec. The representation is not merely compressed imagery; it is an executable, uncertainty-aware causal object whose quality is evaluated by perturbation and behavioral tests.

performance (Google, 2026).

Anthropic’s September 2026 “When AI builds itself” report states that a growing share of AI development is being delegated to AI systems while explicitly stating that fully autonomous recursive self-improvement has not been reached and is not inevitable (Anthropic, 2026). This is evidence for an emerging AI-R&D feedback loop, not proof of a self-sustaining intelligence explosion.

For this paper, AGI and RSI dates are therefore treated as scenario variables. We define an optimistic but deliberately broad AGI planning envelope

$$T_{\text{AGI}} \in [2029, 2035], \quad (31)$$

and a later planning envelope for high-autonomy recursive self-improvement,

$$T_{\text{RSI,AL5}} \in [2035, 2045]. \quad (32)$$

These intervals are not presented as consensus forecasts. They are a planning envelope chosen to stress-test the downstream neuroscience program against a faster AI-development world. The 2025 neuroscientist survey is intentionally retained as a much more conservative counterpoint: its median human-WBE estimate was about 2125 (Zelevnikov-Johnston et al., 2025).

13.1 Operational AGI definition

Because “AGI” lacks a single universally accepted scientific threshold, the present work uses an operational proxy: robust performance at approximately top-expert human levels across essentially all major cognitive task families, together with broad transfer, long-horizon planning, tool use, learning, and self-correction. Passing a benchmark suite alone is insufficient.

13.2 Conditional ASI interval

We model the interval from AGI to early superintelligence as

$$\Delta T_{\text{ASI}} = T_{\text{ASI}} - T_{\text{AGI}}. \quad (33)$$

2026 forecasting work allows fast scenarios in which this interval is measured in months and central scenarios of roughly one to a few years (AI Futures Model, 2026). Accordingly, the Liquid Galactica stress scenario allows early ASI within approximately months to three years after broad AGI. We do not hard-code an instantaneous transition: even highly superhuman cognition may remain constrained by compute availability, experimental throughput, semiconductor production, data access, and deployment governance.

13.3 Why full autonomous RSI is separated from ASI

A system can be superhuman at a fixed capability set without being able to redesign its training stack, acquisition pipeline, evaluation process, infrastructure, and successor-generation process. We therefore reserve “AL5 RSI” for an engineering condition in which an AI system can select improvement directions, design and execute experiments, acquire evidence, modify or train successors, evaluate the result, and recursively repeat the loop without routine human strategic direction. The September 2026 RSI preprint explicitly describes a progression from improvement-execution through improvement-strategy, experience acquisition, environment adaptation, and recursive meta-improvement (Duan et al., 2026).

The central coupling hypothesis can be written as

$$\frac{d\mathcal{P}}{dt} = F(A_{\text{AI}}, M_{\text{meas}}, C_{\text{comp}}, B_{\text{bio}}, H_{\text{pres}}), \quad (34)$$

where $\mathcal{P}$ is progress toward causal brain emulation and the remaining terms represent AI-enabled research throughput, measurement, computation, biological understanding, and preservation capability. The function F is unknown; the equation is a systems-design abstraction.

14. From AGI to mind reconstruction: coupling, not substitution

The relevant causal chain is

$$\begin{aligned} \text{AI} &\rightarrow \text{R\&D} \rightarrow \text{measurement} \rightarrow \text{inference}, \\ &\quad \text{simulation} \rightarrow \text{validation}. \end{aligned} \quad (35)$$

This chain creates five feedback loops:

1. AI improves AI algorithms and hardware.
2. AI improves scientific instruments and image analysis.
3. AI improves biological modelling and parameter inference.
4. Better models generate better experimental designs.
5. Better experiments produce better data for the next generation of models.

The overall progress rate can be conceptualized as

$$\frac{d\mathcal{P}}{dt} = F(A_{\text{AI}}, M_{\text{meas}}, C_{\text{compute}}, B_{\text{bio}}, E_{\text{energy}}), \quad (36)$$

where $\mathcal{P}$ is WBE progress and the arguments represent AI capability, measurement technology, computation, biological understanding, and energy/infrastructure.

Table 1: Working late-2026 AI trajectory adopted by Liquid Galactica. The intervals are scenario windows, not predictions.

Period	Scenario stage	Conservative technical interpretation
2026–2028	Agentic research acceleration	Increasing autonomous coding, scientific search, algorithm discovery, experiment design, and evaluation.
2029–2035	Broad AGI planning envelope	Candidate systems capable of robust top-expert performance across most cognitive task families; definition remains operational and contestable.
2030–2038	Conditional early ASI window	Could follow AGI in months to a few years in fast-takeoff scenarios; not guaranteed.
2035–2045	Full-autonomous RSI (AL5) planning envelope	Strategic research autonomy, successor design, experimentation, and recursive evaluation without routine human strategic direction; not established as of 2026.
2030s	AI-accelerated neuroscience and biotechnology	Faster segmentation, connectomics, molecular inference, simulation, theorem proving, and hardware design.
2030s–2040s	Large-scale WBE engineering	Possible movement from component-scale validation toward increasingly comprehensive causal reconstructions if measurement bottlenecks fall.
2040s+	Digital persistence research	Human-scale reconstruction, if achieved, would require independent validation of memory, dynamics, embodiment, identity, consciousness, and safety.

No single variable controls the future. A rapid AI advance can still encounter a microscope bottleneck; an exceptional microscope can still produce data that no model can interpret; and a perfect structural map can still fail to resolve the physics of memory or consciousness.

15. A biological route to persistence

Digital persistence is not the only path to longer-lived minds. Modern aging research identifies interacting mechanisms including genomic instability, epigenetic alterations, loss of proteostasis, mitochondrial dysfunction, cellular senescence, stem-cell exhaustion, inflammation, and other processes (Lopéz-Otín et al., 2023). Experimental work on partial reprogramming has shown age-associated changes in neuronal and regenerative environments in animals, but this should not be interpreted as demonstrated human biological immortality (Xu et al., 2024).

The research space can therefore be represented by two coupled trajectories:

$$\mathcal{T}_{\text{bio}} = \text{repair the biological substrate}, \quad (37)$$

and

$$\mathcal{T}_{\text{dig}} = \text{preserve} + \text{reconstruct information}. \quad (38)$$

A third hybrid route is possible:

$$\mathcal{T}_{\text{hyb}} = \text{biological repair} + \text{neural replacement} + \text{digital preservation}. \quad (39)$$

This paper does not assert that any of the three routes will succeed. It argues that they should be studied as partially coupled engineering pathways.

16. Preservation before reconstruction

One strategy for reducing the measurement problem is to separate time of death from time of scanning. Aldehyde-stabilized cryopreservation work demonstrated preservation of fine neural ultrastructure in large-mammal brains, and later research has continued to study whole-brain preservation and cryoprotectant transport (Song and Fahy, 2015; Garrood et al., 2026).

The logic is straightforward:

$$\text{preserve } I(t_0) \rightarrow \text{measure } I(t_1), \quad t_1 > t_0. \quad (40)$$

The key scientific limitation is equally straightforward:

$$\text{preserved structure} \not\Rightarrow \text{validated recoverable mind}. \quad (41)$$

The preservation pathway is therefore an information-availability strategy, not a demonstrated resurrection technology.

17. Consciousness, identity, and persistence

The engineering question and the metaphysical question should remain distinct.

A digital reconstruction can be evaluated at several levels:

1. **Behavioral equivalence:** similar observable responses under defined environments.
2. **Functional equivalence:** similar internal computational functions.
3. **Causal equivalence:** comparable response to interventions at relevant scales.
4. **Phenomenal equivalence:** comparable subjective experience.

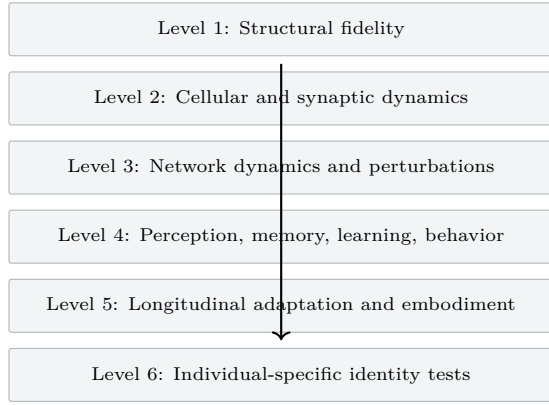


Figure 2: Proposed validation hierarchy. Passing an upper layer does not automatically establish the layers below it.

5. **Personal continuity:** whether the original person’s conscious process continues.

The first three are increasingly approachable as empirical science. The last two remain unresolved questions in philosophy of mind and consciousness science. Contemporary consciousness research contains multiple major theories rather than a single settled mechanism (Consciousness perspective, 2024; Namazi, 2026).

A conservative future protocol should therefore avoid treating a conversationally convincing digital agent as evidence of continuity. Instead, it should pre-register physiological, behavioral, memory, perturbational, and learning criteria before testing the reconstruction.

18. A proposed validation stack

We propose that WBE claims should be graded according to the strongest validated level rather than a binary “uploaded/not uploaded” label.

At each level, a control set should be maintained. For example, a model that reproduces a person’s memory of an object but fails to reproduce how that memory changes after interference should not receive full memory fidelity credit.

18.1 Counterfactual testing

The most informative experiments will likely be counterfactual. Given an intervention I , compare biological and digital outcome distributions:

$$\Delta(I) = D_W(P_B(Y \mid \text{do}(I)), P_R(Y \mid \text{do}(I))). \quad (42)$$

A model that matches observed history but fails intervention tests may be an observational mimic rather than a causal reconstruction.

18.2 Longitudinal identity tests

For individual-specific reconstruction, evaluation should be longitudinal. Let q_t be a vector of individualized behavioral and mnemonic observables. Then

$$E_{\text{id}} = \frac{1}{T} \sum_{t=1}^T \|q_t^{\text{bio}} - q_t^{\text{dig}}\|_W. \quad (43)$$

The metric is not a proof of identity. It is an engineering statistic that measures persistence of individual-specific dynamics.

19. A research architecture for Liquid Galactica

The proposed program is organized around seven technical workstreams.

19.1 Workstream A: measurement

Develop multimodal measurement strategies that can co-register structure, activity, cell type, synaptic properties, molecular state, and body interaction. A key requirement is calibrated uncertainty and provenance for every measurement channel.

19.2 Workstream B: automated reconstruction

Use AI systems to segment, trace, classify, and proofread neural tissue. Active-learning systems should identify regions where additional measurement would most reduce uncertainty rather than processing every voxel uniformly.

19.3 Workstream C: latent state inference

Infer unobserved dynamic variables from partial recordings using state-space models, Bayesian inference, neural operators, and mechanistic priors.

19.4 Workstream D: causal compression

Search for representations that preserve intervention response. Candidate tools include graph sparsification, low-rank approximation, tensor networks, operator learning, model reduction, and learned generative codecs.

19.5 Workstream E: executable substrate

Map the compressed representation onto heterogeneous hardware. Compare GPUs, CPUs, neuromorphic devices, memory-centric accelerators, analog or mixed-signal devices, and future quantum coprocessors where justified by measured workloads.

19.6 Workstream F: validation

Establish a pre-registered battery of structural, physiological, behavioral, mnemonic, perturbational, and longitudinal tests.

19.7 Workstream G: persistence

Only after the preceding layers are validated should the research program address persistent operation, embodiment, identity, legal status, and long-duration safety.

20. A reference implementation architecture

A future prototype could be represented by the following data structures:

```

BrainManifest
  subject_id
  acquisition_history
  modality_graph
  uncertainty_model
  structural_graph
  cell_catalog
  synapse_catalog
  dynamic_state
  plasticity_model
  
```

neuromodulatory_state
embodied_interface
validation_suite
provenance_hash

The ‘provenance_hash’ is not cosmetic. A future digital brain file would contain highly sensitive individual information. Tamper-evident provenance and capability-limited execution should therefore be treated as core architecture.

20.1 State versioning

A useful abstraction is a versioned state sequence:

$$R_{t+\Delta t} = \Gamma(R_t, U_t, \xi_t), \quad (44)$$

where Γ is the transition operator and ξ_t contains stochasticity or uncertainty. Each state can then be addressed by a cryptographic digest and linked to the previous state.

This transforms the future brain emulator from a static file into a versioned computational history.

21. Security and failure modes

A digital human representation would be fundamentally different from an ordinary model checkpoint. It could encode intimate memories, preferences, behavioral vulnerabilities, and possibly mechanisms that reproduce the individual’s identity-specific behavior.

Security requirements would therefore include:

- cryptographic integrity and authenticated state transitions;
- hardware-backed isolation for execution;
- access control at memory, tool, and network layers;
- reversible checkpointing and safe rollback;
- provenance of every reconstruction stage;
- independent validation before any high-autonomy deployment.

A particularly important principle is that a digital brain should not automatically inherit unrestricted access to the internet, financial systems, physical actuators, or other external resources. Capability boundaries should remain independent of the model’s internal fidelity.

22. Experimental ladder: what can be tested now

The entire roadmap does not require waiting for a human WBE system. Each conceptual component yields smaller experiments.

The ladder deliberately prevents an enormous final problem from becoming unfalsifiable. Each stage reduces uncertainty for the next one.

23. Mathematical research agenda

The following mathematical problems are especially relevant.

23.1 Problem 1: causal minimum description

Find the minimum executable representation R satisfying the causal rate-distortion constraint in Equation

(13). Determine conditions under which R is unique, approximately unique, or inherently non-identifiable.

23.2 Problem 2: multiscale state observability

Given partial observations $y(t)$, determine whether the latent state $z(t)$ is observable under a physically realistic measurement operator. For nonlinear systems, characterize the minimal measurement set needed to reconstruct relevant dynamical manifolds.

23.3 Problem 3: uncertainty amplification

Quantify how microscopic uncertainty propagates into macroscopic behavior:

$$\Sigma_T = J_T \Sigma_0 J_T^\top + \Sigma_{\text{process}}, \quad (45)$$

where J_T is the sensitivity operator over horizon T . Regions with high amplification can be prioritized for better measurement.

23.4 Problem 4: causal graph reduction

Given a dynamic graph G and a target intervention family $\mathcal{I}$, find a reduced graph G' that preserves interventions within tolerance. This is related to graph sparsification but with dynamic and causal objectives rather than purely spectral ones.

23.5 Problem 5: quantum-classical boundary

For a biological system with candidate quantum state ρ_Q coupled to classical state z_C , derive a reduced model

$$\dot{\rho}_Q = \mathcal{L}(\rho_Q, z_C), \quad \dot{z}_C = F(z_C, \text{Tr}(O\rho_Q)), \quad (46)$$

that reproduces experimentally observable effects while avoiding unnecessary full quantum simulation.

23.6 Problem 6: energy-aware representation

Optimize the representation jointly for fidelity and energy:

$$\min_R D_c(B, R) + \lambda E(R). \quad (47)$$

A successful solution would make representation learning itself an energy-aware scientific objective.

24. Possible quantum mathematical extensions

24.1 Quantum channel view of measurement

A measurement process can be represented abstractly as a quantum channel

$$\mathcal{N} : \mathcal{L}(\mathcal{H}_B) \rightarrow \mathcal{L}(\mathcal{H}_M), \quad (48)$$

with the measured state $\rho_M = \mathcal{N}(\rho_B)$. If only a partial subsystem is physically accessible, the reconstruction becomes an inverse-channel problem.

The no-cloning theorem places an important conceptual constraint on arbitrary unknown quantum states. A future practical brain scan therefore cannot be assumed to recover an exact hidden quantum state by classical measurement alone. The relevant question would be which quantum variables are actually embodied in biological computation and which can be reduced to classical sufficient statistics.

Table 2: Example experimental ladder for progressively harder causal-emulation tests.

Stage	System	Primary target	Example success criterion
I	Ion-channel / synapse models	Reduced-order dynamics	Reproduce perturbation-response curves within pre-registered tolerance.
II	Small biological circuits	Connectivity + activity	Predict unseen responses to patterned stimulation.
III	Invertebrate nervous systems	Whole-system reconstruction	Reproduce behavior under closed-loop environmental perturbations.
IV	Mammalian cortical volumes	Structure + function	Predict neuronal response from joint structural/functional data.
V	Large mammalian networks	Multiscale dynamics	Preserve activity statistics, perturbational signatures, and learning.
VI	Individual human cortical systems	Personalized model	Reproduce individual-specific memory and behavior under constrained tests.
VII	Human-scale WBE	Full target suite	Causal and longitudinal equivalence across the pre-registered task family.

24.2 Quantum relative entropy

For two candidate quantum representations, one can use

$$S(\rho||\sigma) = \text{Tr} [\rho(\log \rho - \log \sigma)] \quad (49)$$

as an information-geometric discrepancy when the required assumptions are met. A quantum-aware reconstruction pipeline could optimize a weighted combination of classical causal distortion and quantum statistical distance.

24.3 Quantum error correction as a metaphor and an algorithmic tool

Error-correcting codes protect logical information from noisy physical channels. Neural systems are also noisy and adaptive, but there is no basis for claiming that the brain implements a known quantum error-correcting code. Nevertheless, the mathematics of redundancy and protected subspaces may inspire algorithms for identifying robust information-bearing motifs in neural tissue.

The research hypothesis is therefore:

$$\text{biological redundancy} \stackrel{?}{\sim} \text{error-correcting redundancy}, \quad (50)$$

not as an identity, but as a testable information-theoretic analogy.

25. Biophysical state dynamics across scales

A credible WBE substrate should admit multiple mechanistic resolutions. A conductance-based neuron can be written as

$$C_m \dot{V}_i = - \sum_k g_{ik}(V_i - E_{ik}) - I_i^{\text{syn}} + I_i^{\text{ext}} + \xi_i(t), \quad (51)$$

where C_m is membrane capacitance, g_{ik} is the conductance of channel family k , E_{ik} its reversal potential, and ξ_i a stochastic contribution. The Hodgkin–Huxley system provides one classical mechanistic instance (Hodgkin and Huxley, 1952). A reduced model is admissible only when the eliminated variables do

not change the chosen causal observables beyond a pre-specified tolerance.

Synaptic state can be represented as

$$\begin{aligned} I_{ij}^{\text{syn}} &= g_{ij}s_{ij}(t)(V_i - E_{ij}), \\ \dot{s}_{ij} &= \alpha_{ij}T_j(1 - s_{ij}) - \beta_{ij}s_{ij}. \end{aligned} \quad (52)$$

where T_j denotes transmitter-release state. Plasticity introduces additional timescales; a pair-based spike-timing rule is

$$\Delta w_{ij} = \begin{cases} A_+ e^{-\Delta t/\tau_+}, & \Delta t > 0, \\ -A_- e^{\Delta t/\tau_-}, & \Delta t < 0, \end{cases} \quad (53)$$

with $\Delta t = t_{\text{post}} - t_{\text{pre}}$. This is a model family rather than a universal synaptic law.

25.1 Molecular state

For a biochemical subsystem with concentration vector $x \in \mathbb{R}_+^m$,

$$\dot{x} = Nv(x, k, u), \quad (54)$$

where N is a stoichiometric matrix and v contains reaction rates. When molecular copy numbers are small, the chemical master equation is more appropriate:

$$\dot{P}(x, t) = \sum_r [a_r(x - \nu_r)P(x - \nu_r, t) - a_r(x)P(x, t)]. \quad (55)$$

The WBE requirement is not to simulate every reaction by first principles. It is to identify which molecular variables cannot be marginalized without violating the target fidelity criterion.

25.2 Continuum limits

At sufficiently coarse scale, interacting populations can be represented by neural fields. A generic Wilson–Cowan form is

$$\tau_E \dot{E} = -E + \sigma(w_{EE} * E - w_{EI} * I + P_E), \quad (56)$$

$$\tau_I \dot{I} = -I + \sigma(w_{IE} * E - w_{II} * I + P_I), \quad (57)$$

where $*$ denotes spatial convolution. A continuum representation is useful only if the observables of interest are invariant, within tolerance, under this coarse-graining.

26. Information geometry, causal rate-distortion, and sufficient statistics

Ordinary reconstruction loss is not enough for WBE. Let X denote a biological system, R a digital representation, A an intervention, and Y an outcome. Define

$$D_{\text{causal}}(X, R) = \mathbb{E}_{A \sim \pi} \left[D_{KL}(P_X(Y | \text{do}(A)) \| P_R(Y | \text{do}(A))) \right]. \quad (58)$$

A causal rate-distortion problem is then

$$R_C(\varepsilon) = \inf_{p(r|x)} I(X; R) \quad \text{s.t.} \quad D_{\text{causal}}(X, R) \leq \varepsilon. \quad (59)$$

This formalizes the intuition that one should preserve the smallest amount of information that remains predictive under interventions.

For a process Y , predictive information can be written as

$$I_{\text{pred}}(t) = I(Y_{1:t}; Y_{t+1:\infty}). \quad (60)$$

Computational mechanics relates this idea to causal states: pasts that induce the same conditional distribution over futures can be represented by the same state (Crutchfield and Young, 1989). Liquid Galactica proposes an intervention-aware extension in which future equivalence is conditioned not only on passive observation but on a controlled test family.

26.1 Model selection and minimum description length

If R_1 and R_2 both satisfy the causal tolerance, model selection can favor a compact description using an objective such as

$$\mathcal{J}(R) = \underbrace{D_{\text{causal}}(X, R)}_{\text{causal error}} + \lambda \underbrace{L(R)}_{\text{description length}} + \mu \underbrace{E(R)}_{\text{execution energy}}. \quad (61)$$

This produces a three-way tradeoff among fidelity, storage/complexity, and energy rather than a single compression ratio.

27. Graph observability, controllability, and experiment selection

Let a simplified network model be

$$\dot{x}(t) = Ax(t) + Bu(t), \quad y(t) = Cx(t). \quad (62)$$

The finite-horizon controllability Gramian is

$$W_c(T) = \int_0^T e^{At} B B^\top e^{A^\top t} dt, \quad (63)$$

and the observability Gramian is

$$W_o(T) = \int_0^T e^{A^\top t} C^\top C e^{At} dt. \quad (64)$$

Directions corresponding to small eigenvalues are weakly controllable or observable. For brain mapping, this suggests a quantitative principle: allocate additional

measurement to variables whose uncertainty has high downstream causal impact.

For nonlinear dynamics with flow $\Phi_t(z_0)$,

$$J(t) = \frac{\partial \Phi_t}{\partial z_0} \quad (65)$$

propagates local uncertainty approximately as

$$\Sigma_t \approx J(t) \Sigma_0 J(t)^\top + \Sigma_{\text{proc}}(t). \quad (66)$$

A measurement action a can be ranked by expected information gain

$$\text{EIG}(a) = I(\Theta; Y_a | D), \quad (67)$$

normalized by acquisition cost. The result is an active-scanning policy rather than a uniform-volume policy.

28. Hybrid classical–quantum state-space formulation

The quantum part of Liquid Galactica is conditional on experimental evidence. If a biologically relevant quantum subsystem is identified, represent the state as

$$\mathcal{X}(t) = (z_C(t), \rho_Q(t)), \quad (68)$$

where z_C is a classical neural state and ρ_Q is a density operator.

A coupled model can be written

$$\dot{z}_C = F_C(z_C, \text{Tr}(O_1 \rho_Q), \dots, u), \quad (69)$$

$$\dot{\rho}_Q = \mathcal{L}_{z_C}[\rho_Q], \quad (70)$$

with a Lindblad–GKLS generator

$$\mathcal{L}[\rho] = -\frac{i}{\hbar} [H, \rho] + \sum_k \gamma_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\} \right). \quad (71)$$

The important methodological point is that quantum variables enter the model only when they explain independently measured biological effects.

28.1 Quantum model comparison

Let $\mathcal{M}_C$ be a flexible classical model and $\mathcal{M}_Q$ a quantum-aware model. A meaningful quantum claim requires that $\mathcal{M}_Q$ predict data inaccessible to the classical model under matched complexity penalties. Bayesian model comparison, held-out likelihood, causal intervention tests, and independent physical measurement can provide such discrimination.

A critical review published in 2026 emphasizes this requirement: quantum consciousness proposals should be evaluated for physical feasibility in warm biological tissue, explanatory sufficiency, and empirical testability (Frontiers review, 2026). Liquid Galactica therefore treats quantum consciousness as an E4 hypothesis, not a premise.

29. Quantum information and tensor-network compression

For a high-order state tensor $T_{i_1 \dots i_n}$, a tensor-train approximation takes the form

$$T_{i_1 \dots i_n} \approx G_1^{(i_1)} G_2^{(i_2)} \dots G_n^{(i_n)}. \quad (72)$$

For physical dimension d and maximal rank r , the parameter count is polynomial in n, d, r rather than the full tensor size when the ranks remain small. The WBE question is empirical: do the correlation structures relevant to target cognition admit low-rank or sparse representations?

For a quantum subsystem, von Neumann entropy is

$$S(\rho) = -\text{Tr}(\rho \log \rho), \quad (73)$$

and mutual information is

$$I(A : B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB}). \quad (74)$$

These quantities could, in principle, characterize correlations that must be retained. They do not imply that the full brain is quantum computational hardware.

For quantum states ρ and σ , quantum relative entropy is

$$S(\rho \parallel \sigma) = \text{Tr}[\rho(\log \rho - \log \sigma)], \quad (75)$$

and Uhlmann fidelity is

$$F(\rho, \sigma) = \left(\text{Tr} \sqrt{\sqrt{\rho} \sigma \sqrt{\rho}} \right)^2. \quad (76)$$

A future quantum-aware codec could use these only on subsystems for which quantum degrees of freedom are experimentally established (Nielsen and Chuang, 2010).

30. Thermodynamics, information, and the 20-W benchmark

Let total emulator power be

$$P_{tot} = P_{mem} + P_{comm} + P_{comp} + P_{sense} + P_{cool} + P_{aux}. \quad (77)$$

Over an interval $[0, T]$,

$$\mathcal{E}_T = \int_0^T P_{tot}(t) dt. \quad (78)$$

Landauer's lower bound for logically irreversible erasure is

$$E_L = k_B T \ln 2, \quad (79)$$

but this is not a measurement of brain energy or of biological bit erasure. Biological energy expenditure includes ionic gradients, molecular synthesis, transport, synaptic maintenance, signaling, vascular support, and heat dissipation (Landauer, 1961; Attwell and Laughlin, 2001).

Define causal-energy efficiency

$$\eta_C = \frac{\mathcal{U}_C}{\mathcal{E}_T}, \quad (80)$$

where $\mathcal{U}_C$ is a normalized utility derived from a pre-registered intervention suite. This metric allows conventional GPUs, CPUs, neuromorphic devices, analog systems, photonic systems, and future quantum co-processors to be compared using the same causal workload rather than FLOPS alone.

31. Numerical stability and stochastic equivalence

An emulator can be locally accurate yet diverge over long horizons. Let the numerical transition be

$$z_{n+1} = \Psi_{\Delta t}(z_n; \theta), \quad (81)$$

and the target transition be $\Phi_{\Delta t}$. The one-step local error is

$$e_n = \Psi_{\Delta t}(z_n; \theta) - \Phi_{\Delta t}(z_n; \theta). \quad (82)$$

When sensitivity is high, perturbations can grow approximately as

$$\|\delta z(t)\| \sim \|\delta z(0)\| e^{\lambda_{max} t}, \quad (83)$$

where λ_{max} is the maximal Lyapunov exponent of the relevant reduced system. The equation is a generic dynamical-systems relation, not a claim that the brain belongs to a specific chaotic universality class.

The reconstruction should therefore report separate tolerances

$$\varepsilon_{struct}, \varepsilon_{cell}, \varepsilon_{syn}, \varepsilon_{network}, \varepsilon_{memory}, \varepsilon_{behavior}, \varepsilon_{long}. \quad (84)$$

The strongest scientific claim is bounded by the weakest independently validated tolerance.

32. Identity as a dynamical trajectory

If a digital reconstruction is initialized at t_0 , its identity-relevant object is not a single checkpoint but a stochastic trajectory. Let $z_B(t)$ and $z_D(t)$ denote biological and digital states. Define

$$\mathcal{D}_T = \frac{1}{T} \int_0^T \|h_B(z_B(t)) - h_D(z_D(t))\|_W^2 dt, \quad (85)$$

where h extracts experimentally specified observables.

This quantity can measure individual-specific dynamical agreement but cannot, by itself, prove metaphysical identity. After copying, two trajectories $z_D^{(1)}$ and $z_D^{(2)}$ may diverge under different environments or internal noise. The information representation therefore naturally supports branching histories even before philosophical questions of personhood are considered.

33. The digital persistence hypothesis

Liquid Galactica uses the following conditional hypothesis:

If the cognitive and mnemonic properties relevant to an individual can be represented by a sufficiently complete causal state, and if that state can be executed in a substrate that preserves the relevant dynamics and embodied interactions, then a persistent digital instantiation of those properties may become physically realizable.

The statement contains several conditional clauses because each corresponds to an unresolved scientific requirement.

We define a digital persistence chain:

$$P = M \circ I \circ R \circ V \circ E, \quad (86)$$

where E is encoding, V validation, R reconstruction, I initialization of dynamic state, and M maintenance under long-duration execution.

Failure of any stage can break persistence.

34. Why “digital immortality” must remain a hypothesis

The phrase “digital immortality” is useful as a long-term human objective, but it should not be treated as a scientific observation. Even a perfect behavioral duplicate would not automatically settle whether a first-person conscious perspective persists through copying.

For this reason, the proposed terminology is:

- **digital preservation** for storing an individual’s neural information;
- **digital reconstruction** for building an executable model;
- **digital persistence** for long-duration operation of that model;
- **digital continuity** only for cases where continuity criteria have been explicitly established;
- **digital immortality** as a broader philosophical and societal objective rather than a current scientific result.

This vocabulary keeps engineering claims separate from unresolved philosophy.

35. A systems view of the next decades

The central scenario of Liquid Galactica can be summarized as five coupled technological curves:

$$\{A(t), D(t), C(t), B(t), H(t)\} \quad (87)$$

where A is AI capability, D measurement density, C computational efficiency, B biological knowledge, and H preservation/handling technology.

A qualitative phase diagram is useful:

1. **Phase I — digitize portions:** large neural datasets and component models.
2. **Phase II — integrate modalities:** structure, function, chemistry, and behavior become co-registered.
3. **Phase III — causal emulation:** models reproduce intervention response rather than only observations.
4. **Phase IV — personalized reconstruction:** individual-specific state and memory become reconstructable.
5. **Phase V — persistent execution:** digital systems maintain their dynamics over long periods and interact with environments.

The timeline to Phase IV or V is currently unknown. The optimistic scenario used here places a major AI acceleration in the late 2020s and expects its scientific consequences to propagate during the 2030s and beyond. The 2025 neuroscientist survey is much more conservative on human WBE dates, illustrating the unusually large uncertainty between expert opinion and optimistic technology scenarios (Zelezniak-Johnston et al., 2025). Both perspectives belong in the literature review.

36. Discussion

The strongest evidence available today is distributed across different domains. Connectomics demonstrates that large neural structures can be digitized at extraordinary resolution. Large-scale simulation demonstrates that neuron-count-scale networks can be executed on real hardware. Functional connectomics joins structure and activity. Memory research establishes that individual information depends on distributed biological mechanisms. Brain-inspired computing demonstrates the importance of local memory, event-driven computation, and energy efficiency. Quantum biology demonstrates that quantum mechanisms can matter in specific biological processes without implying a quantum theory of consciousness. AI research systems demonstrate that portions of scientific reasoning can be increasingly automated.

No single result solves the mind-uploading problem. The research opportunity arises from their convergence.

The deepest engineering question is perhaps therefore a compression problem:

How much physical detail can be removed while preserving causal mind-level behavior?

(88)

A future answer could reduce the raw zettabyte-scale imaging problem to a structured representation that is far smaller and more executable. It might also reveal that certain “microscopic” details are irrelevant at the cognitive timescale, while others are disproportionately important.

This is why the proposed framework is not committed to any single neuron model, hardware platform, or quantum theory. It is a hierarchy of tests.

37. Limitations

First, the paper does not provide new experimental evidence. All quantitative claims are literature-derived or explicitly marked as projections. Second, the minimum sufficient brain state is not known. Third, current human-scale digital brain simulations are not individualized mind uploads. Fourth, no established experiment currently resolves whether a digital reconstruction would preserve subjective consciousness or personal identity. Fifth, raw storage extrapolations depend strongly on imaging resolution, compression, file representation, and what is retained. Sixth, AGI and ASI dates are intrinsically uncertain and should not be treated as deterministic engineering milestones. Seventh, quantum formulations in this paper are optional modelling branches, not evidence that cognition requires quantum coherence or that the brain is a quantum computer.

38. Research agenda and recommended near-term program

The next decade should prioritize measurable bottlenecks rather than philosophical endpoints.

1. Build increasingly complete multimodal datasets that align structure, activity, chemistry, and behavior in the same specimen.

2. Develop uncertainty-aware neural reconstruction systems that can quantify confidence at the synapse and cell level.
3. Create causal benchmarks in which models must predict responses to interventions that were excluded from training.
4. Develop compression algorithms that optimize causal fidelity rather than pixel reconstruction error.
5. Compare brain-inspired, neuromorphic, memory-centric, and conventional accelerators on identical WBE workloads.
6. Develop mathematical criteria for identifying when microscopic degrees of freedom can be safely marginalized.
7. Maintain a quantum-biology branch in which candidate quantum mechanisms are tested experimentally rather than assumed.
8. Establish preservation standards that retain future informational value without claiming current reversibility.
9. Build an open benchmark suite for memory, plasticity, perturbation response, and longitudinal individual-specific dynamics.
10. Treat security, privacy, consent, and execution isolation as first-class technical requirements from the beginning.

39. Conclusion

The Liquid Galactica program starts from a simple observation: the human brain is both enormously complex and remarkably efficient. Its approximately 20-W power budget, massive parallelism, continual plasticity, and distributed memory mechanisms pose a challenge to conventional computing and an opportunity for a new generation of neurocomputational architectures.

Recent work on digital-twin brains similarly argues that model fidelity is fundamentally constrained by the resolution, completeness, and updatability of the measurements rather than by neuron count alone (Zanichelli et al., 2025). Current science already provides several pieces of the future puzzle. Human cortex can be reconstructed at petascale from tiny tissue volumes. Mouse experiments increasingly integrate structure and function. Brain-scale simulations now reach tens of billions of model neurons. AI systems can already automate parts of scientific discovery, algorithm development, coding, and research synthesis. Frontier AI is simultaneously becoming a potentially stronger amplifier of the scientific process.

The path from these achievements to a persistent digital mind is still long and uncertain. The missing problem is not merely storage. It is the identification and preservation of the causal information that makes an individual brain behave as that individual brain.

Liquid Galactica proposes to attack this problem as a unified engineering and mathematical discipline:

$$\begin{array}{l} \text{Measure} \rightarrow \text{Infer} \rightarrow \text{Represent} \rightarrow \text{Compress} \\ \text{Execute} \rightarrow \text{Perturb} \rightarrow \text{Validate} \rightarrow \text{Preserve} \end{array} \quad (89)$$

Table 3: Core notation for the Liquid Galactica framework.

Symbol	Definition
$\mathcal{B}(t)$	Multiscale physical brain state at time t
R	Digital causal representation
D_{raw}	Raw measurement data volume
D_c	Task-conditioned causal distortion
$\mathcal{T}$	Validation-test family
$z(t)$	Latent dynamical state
$\theta(t)$	Time-varying model parameters
ρ	Density operator for a quantum subsystem
$\mathcal{L}$	Generator of open-system quantum dynamics
$\mathcal{C}$	Causal reconstruction operator
E	Energy expenditure / budget

The long-term human objective can be stated carefully. Humanity may eventually seek forms of persistence that are not restricted to the lifespan of a biological body. Biological rejuvenation, neural replacement, preservation, and digital reconstruction may each contribute to that objective. Whether any one of them eventually supports consciousness or personal continuity is not known.

What is knowable is the sequence of engineering questions.

What information does a brain contain? Which of it matters? Which of it can be measured? Which of it can be compressed?

Which dynamics must be reproduced? Which substrate can execute them? Which tests can distinguish a causal reconstruction from a behavioral imitation?

These questions are sufficiently concrete to begin now.

A. Notation, units, and scaling identities

Table 3 collects the principal variables used throughout the framework. We deliberately distinguish storage, communication, and computation: a representation with small byte count can still induce large bandwidth or recomputation cost.

For an isotropic voxel representation,

$$D_{\text{raw}} \propto r^{-3}. \quad (90)$$

Thus a twofold improvement in linear resolution increases raw voxel count by a factor of eight. This cubic scaling is one reason algorithmic representations, sparse measurements, adaptive sampling, and learned reconstruction are likely to be essential.

B. Measurement-to-emulation accounting

A useful future accounting identity is

$$D_{\text{total}} = D_{\text{img}} + D_{\text{state}} + D_{\text{model}} + D_{\text{uncert}}, \quad (91)$$

where the terms represent imaging archives, instantaneous state variables, executable model parameters, and uncertainty representations. The sum is not necessarily minimal because some terms can be regenerated algorithmically from other terms.

Table 4: Representative information scales.

Scale	Representative quantity
H01 fragment	~ 1.4 PB image data; ~ 150 M synapses
Human, 25 nm	~ 128 – 180 EB raw imaging estimate
Human, 10 nm	~ 2 – 2.8 ZB raw imaging estimate
Human-scale digital model	86 B model neurons in one published simulation
Human brain power benchmark	~ 20 W

The H01 human-cortex reconstruction provides an empirical anchor: approximately 1.4 PB for a cubic-millimetre-scale fragment containing roughly 150 million synapses (Shapson-Coe et al., 2024). The State of Brain Emulation Report estimates approximately 128–180 EB for a 25-nm whole-human-brain regime and approximately 2–2.8 ZB at 10 nm under its stated assumptions (Zanichelli et al., 2025). The discrepancy between raw archive size and executable model size should remain explicit in future systems studies.

C. Forecast provenance and uncertainty

Forecasts are treated as scenario inputs rather than experimental constants. The August 2026 AI Futures Model places substantial probability mass on late-2020s/early-2030s top-expert-level AI in several model configurations while maintaining wide distributions for superintelligence (AI Futures Model, 2026). Metaculus aggregates a different population and question formulation; its current community forecasts should therefore not be interpreted as an expert consensus (Metaculus, 2026). The International AI Safety Report 2026 likewise emphasizes rapid capability growth alongside severe uncertainty in long-horizon autonomy and future trajectories (International AI Safety Report, 2026).

Liquid Galactica consequently uses a three-scenario AI environment,

$$\mathcal{S}_{AI} = \{\text{rapid, intermediate, slow}\}, \quad (92)$$

with the optimistic planning envelope

$$T_{\text{AGI}} \in [2029, 2035], \quad T_{\text{RSI,AL5}} \in [2035, 2045], \quad (93)$$

and a conditional ASI interval of months to a few years after broad AGI in faster scenarios. These are stress-test assumptions, not predictions of guaranteed events.

D. Quantum research matrix

The quantum branch must preserve a strict evidence hierarchy. Quantum effects are experimentally established in several biological settings, including electron-transfer, photosynthetic energy transfer, radical-pair chemistry, and spin-dependent phenomena (Lambert et al., 2021; Hore and Mouritsen, 2016; Kim et al., 2023). These observations do not establish that the brain is a quantum computer, nor that consciousness requires macroscopic quantum coherence.

A useful hypothesis ladder is:

Table 5: Final scenario envelopes used in the paper.

Period	Working interpretation
2026–2028	Agentic coding, scientific assistance, algorithm discovery
2029–2035	Optimistic broad AGI planning envelope
2030–2038	Conditional early ASI window in faster scenarios
2035–2045	Full-autonomous RSI (AL5) planning envelope
2030s	Potential AI-accelerated neuroscience and longevity
2040s+	Potential human-scale reconstruction / digital persistence research

1. **Established:** quantum mechanics governs molecular and electronic processes in biological matter.
2. **Plausible modeling tool:** open-system quantum formalisms can model selected microscopic neural chemistry.
3. **Testable hypothesis:** quantum variables improve prediction of a measurable neural phenomenon beyond classical models.
4. **Speculative:** quantum coherence or quantum-information integration is necessary for consciousness.

The fourth category includes proposals such as Orch OR and related microtubule hypotheses (Hameroff and Penrose, 2014; Hameroff, 2025). Such proposals are useful as falsifiable research prompts, but they are not treated here as established mechanisms. Quantum-cognition models can similarly be useful mathematically without implying physical quantum processing in the brain (Namazi, 2026).

E. Minimum experimental program

A conservative empirical program can proceed without assuming quantum consciousness or immediate human uploading. First, build benchmark regions for which dense structural and functional data are available. Second, learn posterior distributions over latent parameters rather than point estimates. Third, test whether coarse-grained representations preserve intervention responses. Fourth, compare classical, neuromorphic, and quantum-informed models using equalized tasks and energy budgets.

For a model family $\mathcal{M}$ and intervention set $\mathcal{I}$, define

$$\mathcal{E}(\mathcal{M}) = \mathbb{E}_{i \sim \mathcal{I}} [d(y_i^{\text{bio}}, y_i^{\text{model}})], \quad (94)$$

and require uncertainty-calibrated intervals around the model outputs. A useful successor metric is energy-normalized predictive accuracy,

$$\eta_{\text{causal}} = \frac{\text{Predictive utility}}{E_{\text{compute}}}, \quad (95)$$

which connects biological efficiency with emulator design.

Author profile

Devanik Debnath (Devanik21) is an Electronics and Communication Engineering student at a National In-

stitute of Technology in India and an independent technical researcher. His public technical work is available through the Devanik21 GitHub profile (Debnath, 2026). The present manuscript is an independent conceptual and quantitative research preprint; institutional affiliation is not implied beyond the educational descriptor above.

Acknowledgements

This manuscript is a final preprint version of the Liquid Galactica research framework. The literature synthesis intentionally remains conservative with respect to unsettled questions in consciousness, quantum neuroscience, and AI timelines.

References

- A. Sandberg and N. Bostrom. *Whole Brain Emulation: A Roadmap*. Technical Report 2008-3, Future of Humanity Institute, University of Oxford, 2008. <https://ora.ox.ac.uk/objects/uuid:a6880196-34c7-47a0-80f1-74d32ab98788>.
- N. Zanichelli, M. Schons, I. Freeman, P. Shiu, and A. Arkhipov. State of Brain Emulation Report 2025. arXiv:2510.15745, 2025. <https://arxiv.org/abs/2510.15745>.
- A. Shapson-Coe et al. A petavoxel fragment of human cerebral cortex reconstructed at nanoscale resolution. *Science*, 384(6696):eadk4858, 2024. doi:10.1126/science.adk4858.
- C. Lu et al. Simulation and assimilation of the digital human brain. *Nature Computational Science*, 4, 2024. doi:10.1038/s43588-024-00731-3.
- MICrONS Consortium. Functional connectomics spanning multiple areas of mouse visual cortex. *Nature*, 640:435–447, 2025. doi:10.1038/s41586-025-08790-w.
- S. Dorkenwald et al. Neuronal wiring diagram of an adult fruit fly brain. *Nature*, 2024. doi:10.1038/s41586-024-07558-y.
- A. Zelezniak-Johnston, E. F. Kendziorra, and A. T. McKenzie. What are memories made of? A survey of neuroscientists on the structural basis of long-term memory. *PLOS ONE*, 20(6):e0326920, 2025. doi:10.1371/journal.pone.0326920.
- C. López-Otín et al. Hallmarks of aging: An expanding universe. *Cell*, 186(2):243–278, 2023. doi:10.1016/j.cell.2022.11.001.
- X. Xu et al. Restoration of neuronal progenitors by partial reprogramming in the aged neurogenic niche. *Nature Aging*, 2024. doi:10.1038/s43587-024-00594-3.
- Y. C. Song and G. M. Fahy. Aldehyde-stabilized cryopreservation. *Cryobiology*, 71(2):181–197, 2015.
- M. Garrood, A. Keberle, A. Slaughter, A. Sowa, E. L. Thorn, C. De Sanctis et al. Cryopreservation of aldehyde-fixed whole brains. *PLOS ONE*, 21(8):e0344932, 2026. doi:10.1371/journal.pone.0344932.
- H. J. Chiel, J. S. Coggan, G. Datta, J.-M. Fellous, W. R. P. Nourse, R. D. Quinn, P. J. Thomas. Brain-inspired energy efficient technologies for next-generation artificial intelligence. *Biological Cybernetics*, 120:5, 2026. doi:10.1007/s00422-026-01038-4.
- U. Bruno, M. Farronato, A. Salleo, S. Fabiano, F. Santoro. Figures of merit for neuromorphic devices through biological emulation efficacy. *Nature Reviews Electrical Engineering*, 3:579–595, 2026. doi:10.1038/s44287-026-00321-7.
- N. Lambert et al. Quantum biology: An update and perspective. *Quantum Reports*, 3, 2021.
- P. J. Hore and H. Mouritsen. The radical-pair mechanism of magnetoreception. *Annual Review of Biophysics*, 45:299–344, 2016. doi:10.1146/annurev-biophys-032116-094545.
- K. W. Kim et al. Essential elements of radical pair magnetosensitivity in *Drosophila*. *Nature*, 615:111–116, 2023. doi:10.1038/s41586-023-05735-z.
- S. Hameroff and R. Penrose. Consciousness in the universe: A review of the Orch OR theory. *Physics of Life Reviews*, 11(1):39–78, 2014. doi:10.1016/j.plrev.2013.08.002.
- S. Hameroff. A quantum microtubule substrate of consciousness is experimentally supported and solves the binding and epiphenomenalism problems. *Neuroscience of Consciousness*, 2025, niaf011. doi:10.1093/nc/niaf011.
- S. Namazi. Quantum information processing in human cognition: Bridging quantum phenomena, neural dynamics, and decision-making. *Advanced Quantum Technologies*, 2026. doi:10.1002/qute.70327.
- Integrative perspective authors. An integrative, multiscale view on neural theories of consciousness. *Neuron*, 2024.
- J. Gottweis et al. Towards an AI co-scientist. arXiv:2502.18864, 2025. <https://arxiv.org/abs/2502.18864>.
- C. Lu, C. Lu, R. T. Lange, Y. Yamada, S. Hu, J. Foerster, D. Ha, J. Clune. Towards end-to-end automation of AI research. *Nature*, 651:914–919, 2026. doi:10.1038/s41586-026-10265-5.
- AlphaEvolve team. AlphaEvolve: How our Gemini-powered coding agent is scaling impact across fields. Google DeepMind, 2026. <https://deepmind.google/blog/alphaevolve-impact/>.
- Y. Duan et al. The Last AI Built by Humans: Toward Genuine Recursive Self-Improvement. arXiv:2609.11873, 2026. <https://arxiv.org/abs/2609.11873>.
- OpenAI. GPT-6 Astra: A new generation of intelligence. OpenAI, September 2026. <https://openai.com/index/gpt-6-astra/>.
- Anthropic. Introducing Claude Fable 5.1 and Claude Mythos 5.1. Anthropic, September 2026. <https://www.anthropic.com/claude-fable-and-mythos-5-1>.
- Google. Q2 2026 earnings call: Remarks from our CEO. Google, July 2026. <https://blog.google/company-news/inside-google/message-ceo/alphabet-earnings-q2-2026/>.
- International AI Safety Report. International AI Safety Report 2026. 2026. <https://internationalaisafetyreport.org/publication/international-ai-safety-report-2026>.
- AI Futures Model. AI Futures Model: Timelines and takeoff, August 2026 version. 2026. <https://www.aifuturesmodel.com/>.
- Metaculus. AGI forecasting questions and current community aggregates. Accessed September 2026. <https://www.metaculus.com/questions/?search=AGI>.
- E. Romero et al. Quantum coherence in photosynthesis for efficient solar-energy conversion. *Nature Physics*, 10:676–682, 2014.
- R. Landauer. Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5(3):183–191, 1961.
- J. Preskill. Quantum computing in the NISQ era and beyond.

- Quantum*, 2:79, 2018. doi:10.22331/q-2018-08-06-79.
- A. L. Hodgkin and A. F. Huxley. A quantitative description of membrane current and its application to conduction and excitation in nerve. *The Journal of Physiology*, 117(4):500–544, 1952. doi:10.1113/jphysiol.1952.sp004764.
- J. P. Crutchfield and K. Young. Inferring statistical complexity. *Physical Review Letters*, 63(2):105–108, 1989. doi:10.1103/PhysRevLett.63.105.
- D. Attwell and S. B. Laughlin. An energy budget for signaling in the grey matter of the brain. *Journal of Cerebral Blood Flow & Metabolism*, 21(10):1133–1145, 2001. doi:10.1097/00004647-200110000-00001.
- M. A. Nielsen and I. L. Chuang. *Quantum Computation and Quantum Information*. Cambridge University Press, 10th Anniversary Edition, 2010.
- Frontiers in Psychology review. Critical perspectives on quantum theories of consciousness: physical feasibility, explanatory scope, and empirical testability. *Frontiers in Psychology*, 2026. doi:10.3389/fpsyg.2026.1730965.
- Anthropic. When AI builds itself: Measuring the emerging AI–R&D feedback loop. Anthropic, September 2026. <https://www.anthropic.com/institute/recursive-self-improvement>.
- D. Debnath. Devanik21 public technical profile. GitHub, accessed September 20, 2026. <https://github.com/Devanik21>.